Part 3 · Answer Keys & Explanations

Use the quick keys to score fast, then read the full explanations to learn from every question. "CR" marks a constructed-response item (SCR or ECR) scored with a rubric — see the SCR/ECR scored samples.

Quick Answer Keys — Mathematics

#	Test 1	Test 2	Test 3	Test 4	#	Test 1	Test 2	Test 3	Test 4
1	A	A	A	A	21	B	C	B	C
2	B	B	B	B	22	C	B	C	C
3	6.8×10^{4}	4.5×10^{-4}	5.2×10^{6}	7.2×10^{-5}	23	25	41	17	29
4	C	B	A	A	24	A	B	A	C
5	B	B	A	B	25	A	B	B	A
6	B	A	B	A	26	B	B	A	B
7	B	$y=-2x+20$	$y=25x$	$y=15x+40$	27	B	A	A	A
8	$y=3x+6$	B	B	B	28	D	C	C	D
9	B	C	C	A	29	C	B	C	B
10	A	63	40	72	30	C	C	C	C
11	B	B	A	A	31	B	B	78.5	A
12	36	6	5	8	32	D	C	D	C
13	C	A	A	A	33	2	3	2	2
14	C	B	B	B	34	C	C	C	B
15	4	B	B	B	35	B	C	A	C
16	A	B	B	B	36	B	C	C	C
17	A	A	A	A	37	$160	$216	$300	$360
18	B	B	B	B	38	C	B	D	B
19	A,C	A,C	A,C	A,C	39	C	C	C	C
20	C	C	C	C	40	B	B	B	B

Quick Answer Keys — Reading Language Arts

#	T1	T2	T3	T4	#	T1	T2	T3	T4	#	T1	T2	T3	T4
1	A	B	B	B	11	B	B	B	B	21	B	B	B	B
2	B	B	B	B	12	B	B	A	B	22	C	B	B	B
3	C	C	B	B	13	C	B	B	B	23	B	B	B	B
4	B	A	B	B	14	B	B	B	B	24	A	A	A	A
5	B	B	B	A	SCR2	CR	CR	CR	CR	25	A	A	A	A

6	C	C	C	B	15	B	A	A	B	26	A	A	A	A
7	B	B	B	B	16	B	B	B	B	27	A	A	A	D
SCR1	CR	CR	CR	CR	17	B	B	B	B	28	A	A	A	A
8	B	B	B	B	18	B	B	B	B	ECR	CR	CR	CR	CR
9	B	B	B	B	19	B	A	B	B					
10	A	B	B	A	20	B	B	B	B					

The detailed explanations that follow show **why** each answer is correct — and why the common wrong choices are wrong. Read them even for questions you got right.

Mathematics — Answer Explanations

Each explanation gives the correct answer, a worked solution, the common wrong-answer trap, and the TEKS. ★ = Readiness, ○ = Supporting.

Practice Test 1

1. A *(8.2D ★)* — Convert to decimals: ⅔≈0.667, 68%=0.68, 0.7=0.70, √0.5≈0.707 → ⅔, 68%, 0.7, √0.5. *Trap: not evaluating √0.5.*

2. B *(8.2B ○)* — $6^2=36$, $7^2=49$; $36<40<49$ → between 6 and 7.

3. 6.8×10^4 *(8.2C ○)* — 68,000, decimal moved 4 places.

4. C *(8.2A ○)* — π is irrational; √36=6, ⅘, 0.25 are rational.

5. B *(8.4C ★)* — (11−2)/(4−1)=9/3=3. *Trap: D uses rise only.*

6. B *(8.4B ★)* — Unit rate = 12/3 = 4.

7. B *(8.4A ○)* — Similar right triangles → constant rise÷run.

8. $y = 3x + 6$ *(8.5I ★)* — Slope 3, y-intercept 6.

9. B *(8.5G ★)* — Each input one output; A/C/D repeat an x.

10. A *(8.5A ○)* — y=4x passes through the origin; others add a constant.

11. B *(8.5B ○)* — y = 6x + 15. *Trap: A swaps rate and fee.*

12. 36 *(8.5E ○)* — k=20/5=4; 4×9=36.

13. C *(8.5F ○)* — Ratios 4, 3.5, 3.33 not constant.

14. C *(8.5H ○)* — $2/lb, no fee → proportional.

15. 4 *(8.8C ★)* — 4x=16 → x=4.

16. A *(8.8A ○)* — 2x−3 = x+5.

17. A *(8.8B ○)* — 5x+10 = 3x+30.

18. B *(8.9A ○)* — Intersection (2,5): x=2, y=5.

19. A, C *(8.5I ★)* — Slope 2, y-intercept −3.

20. C *(8.7A ★)* — $\pi r^2h = 3.14\times16\times7 = 351.68$.

21. B *(8.7A ★)* — $\frac{1}{3}\pi r^2h = \frac{1}{3}\times3.14\times36\times10 = 376.8$. *Trap: C forgets ⅓.*

22. C *(8.7B ★)* — $48\pi+18\pi = 66\pi \approx 207.24$.

23. 25 *(8.7C ★)* — $49+576 = 625 = 25^2$.

24. A *(8.7D ○)* — Legs 4, 3 → 5.

25. A *(8.3C ★)* — (3×−2, 3×5) = (−6, 15).

26. B *(8.10C ★)* — (x,y)→(−x,y) → (−5, −4).

27. B *(8.10C ★)* — 90° CCW (x,y)→(−y,x) → (−7, 2).

28. D *(8.10A/B ○)* — Dilation changes size; the rest keep congruence.

29. C *(8.8D ○)* — 6x=180, x=30; largest 3x=90°.

30. C *(8.3A ○)* — 6×(5/2)=15.

31. B *(8.6A ○)* — B = base area = πr^2.

32. D *(8.10D ○)* — Area ×k^2: 16×9=144. *Trap: B uses ×3.*

33. 2 *(8.11B ○)* — Mean 6; MAD = (3+1+1+3)/4 = 2.

34. C *(8.5D ★)* — 1.5×8+2 = 14.

35. B *(8.11A ○)* — Falling points → negative linear.

36. B *(8.5C ○)* — Straight upward path → linear.

37. $160 *(8.12D ★)* — 800×0.05×4 = 160.

38. C *(8.12A ○)* — Longer loan → most interest.

39. C *(8.12C ○)* — Regular + compound → grows.

40. B *(8.12G ○)* — (12,000−4,000)/16 = 500.

Practice Test 2

1. A *(8.2D ★)* — ¾=0.75, 0.72, √0.6≈0.775, 78%=0.78 → 0.72, ¾, √0.6, 78%.

2. B *(8.2B ○)* — $81<90<100$ → 9 and 10.

3. 4.5×10^{-4} *(8.2C ○)* — 0.00045, 4 places.

4. B *(8.2A ○)* — √16=4 rational; others irrational.

5. B *(8.4C ★)* — 8/4 = 2.

6. A *(8.4B ★)* — 15/5 = 3.

7. $y = -2x + 20$ *(8.5I ★)* — Start 20, rate −2.

8. B *(8.5G ★)* — Each input once.

9. C *(8.5B ○)* — +4 → non-proportional.

10. 63 *(8.5E ○)* — k=9; 9×7=63.

11. B *(8.5F ○)* — Ratio 4 constant.

12. 6 *(8.8C ★)* — 4x=24 → x=6.

13. A *(8.8A ○)* — 3x+5 = 2x+12.

14. B *(8.9A ○)* — (−3,2): x=−3, y=2.

15. B *(8.4A ○)* — Similar triangles → same slope.

16. B *(8.5H ○)* — Steady 6 mph, no fee → proportional.

17. A *(8.8B ○)* — 4x+6 = 2x+18.

18. B *(8.5B ○)* — y = 3x + 4.

19. A, C *(8.5I ★)* — Slope −3, y-intercept 6.

20. C *(8.7A ★)* — 3.14×25×6 = 471.

21. C *(8.7A ★)* — 4⁄3×3.14×27 = 113.04.

22. B *(8.7B ★)* — 40π+8π = 48π ≈ 150.72.

23. 41 *(8.7C ★)* — 81+1600 = 1681 = 41².

24. B *(8.7D ○)* — √(36+64) = 10.

25. B *(8.3C ★)* — (8, −6).

26. B *(8.10C ★)* — (x,y)→(x,−y) → (−6, −2).

27. A *(8.10C ★)* — 180° → (−4, −1).

28. C *(8.10A/B ○)* — Translation preserves congruence.

29. B *(8.8D ○)* — Corresponding angles equal → 65°.

30. C *(8.3A ○)* — 9×(4/3)=12.

31. B *(8.6C ○)* — $a^2+b^2=c^2$.

32. C *(8.10D ○)* — Area ×k²: 15×4=60.

33. 3 *(8.11B ○)* — Mean 9; MAD = (5+1+1+5)/4 = 3.

34. C *(8.5D ★)* — 3×6+5 = 23.

35. C *(8.11A ○)* — No pattern → no association.

36. C *(8.5C ○)* — U-shape → non-linear.

37. $216 *(8.12D ★)* — 1,200×0.06×3 = 216.

38. B *(8.12A ○)* — Higher rate raises cost.

39. C *(8.12C ○)* — Regular + compound grows most.

40. B *(8.12G ○)* — (16,000−6,000)/20 = 500.

Explanations for Math Tests 3 & 4 continue on the following pages.